

DATA ANALYST COURSE CURRICULUM

Duration: 3 Months (12 Weeks) | Mode: Instructor-Led + Self-Paced + Projects

Tools Covered: SQL | Python | Cloud Computing (Azure) | Azure Data Factory | Power BI

MONTH 1: FOUNDATIONS — SQL & DATA HANDLING

Week 1: Introduction to Data Analytics & SQL Basics

- What is Data Analytics? Role of a Data Analyst
- Data Analytics Lifecycle: Collect → Clean → Analyze → Visualize → Report
- Types of Data: Structured, Semi-structured, Unstructured
- Introduction to RDBMS (MySQL / SQL Server)
- Installing and setting up SQL environment
- Basic SQL: SELECT, WHERE, ORDER BY, DISTINCT, LIMIT

Week 2: Intermediate SQL

- Aggregate Functions: COUNT, SUM, AVG, MIN, MAX
- GROUP BY and HAVING clauses
- Joins: INNER, LEFT, RIGHT, FULL OUTER
- Subqueries and Nested Queries
- Set Operations: UNION, INTERSECT, EXCEPT

Week 3: Advanced SQL

- Window Functions: RANK, ROW_NUMBER, LEAD, LAG
- Common Table Expressions (CTEs) and Views
- Case Statements & Conditional Logic
- Data Cleaning using SQL (handling NULLs, duplicates)
- Query Optimization Basics
- Hands-on: Business Reporting Queries (Sales/HR dataset)

Week 4: SQL Project & Data Modeling Basics

- Star Schema vs Snowflake Schema (Overview)
- Introduction to Data Warehousing Concepts
- OLTP vs OLAP
- Mini Project: End-to-end SQL Case Study (Data extraction + reporting)
- Assessment: SQL Proficiency Test

MONTH 2: PYTHON FOR DATA ANALYSIS & CLOUD FOUNDATIONS

Week 5: Python Programming Basics

- Python Syntax, Variables, Data Types
- Conditional Statements & Loops
- Functions and Modules
- Working with Lists, Tuples, Dictionaries, Sets
- File Handling (CSV, Excel, JSON)

Week 6: Data Analysis with Python (Pandas & NumPy)

- Introduction to NumPy: Arrays and Operations
- Pandas: Series & DataFrames
- Data Cleaning: Missing Values, Duplicates, Outliers
- Data Transformation: Merge, Join, Pivot, GroupBy
- Exploratory Data Analysis (EDA) techniques

Week 7: Data Visualization with Python

- Introduction to Matplotlib and Seaborn
- Creating Charts: Bar, Line, Scatter, Histogram, Box Plot
- Correlation & Trend Analysis
- Storytelling with Data
- Mini Project: EDA + Visualization on a real-world dataset

Week 8: Cloud Computing Fundamentals (Azure)

- Introduction to Cloud Computing: IaaS, PaaS, SaaS
- Overview of Azure Ecosystem
- Azure Storage: Blob Storage, Data Lake Storage
- Azure SQL Database Basics
- Introduction to Azure Portal and Resource Groups
- Data Security & Access Management (Basics)

MONTH 3: AZURE DATA FACTORY, POWER BI & CAPSTONE PROJECT

Week 9: Azure Data Factory (ADF) — Fundamentals

- Introduction to ETL/ELT and Data Pipelines
- Azure Data Factory Architecture & Components
- Linked Services, Datasets, and Pipelines
- Copy Data Activity: Moving data between sources
- Scheduling and Triggers

Week 10: Azure Data Factory — Advanced

- Data Flows: Mapping Data Flows for Transformation
- Integration Runtime concepts
- Monitoring & Debugging Pipelines
- Connecting ADF with Azure SQL, Blob Storage, and Power BI
- Hands-on: Build an End-to-End Data Pipeline using ADF

Week 11: Power BI — Data Visualization & Reporting

- Introduction to Power BI Desktop & Service
- Connecting to Data Sources (SQL, Excel, Azure)
- Data Modeling: Relationships, Star Schema in Power BI
- Power Query: Data Cleaning & Transformation (M Language basics)
- DAX Basics: Calculated Columns & Measures
- Building Interactive Dashboards & Reports
- Publishing & Sharing Reports (Power BI Service)

Week 12: Capstone Project & Interview Preparation

- Real-Time Capstone Project: SQL + Python + Azure Data Factory + Power BI Integration (Extract data → Transform via ADF → Analyze with Python/SQL → Visualize in Power BI)
- Resume Building & Portfolio Development
- Mock Interviews & Commonly Asked Interview Questions
- Doubt-Clearing Session & Final Assessment

LEARNING APPROACH

Component	Description
Self-Learning	Pre-reads, recorded sessions, practice exercises
Instructor-Led Live Sessions	Concept walkthroughs and live coding
Hands-on Labs	Weekly assignments and dataset-based practice
Projects	1 project per month + 1 final capstone project

KEY DELIVERABLES BY COURSE END

- 3 Mini Projects (SQL, Python/EDA, Power BI)
 - 1 Capstone Project (End-to-End Data Pipeline & Dashboard)
 - Resume + Portfolio (GitHub/Power BI published reports)
 - Interview Preparation Kit
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TOOLS COVERED

SQL (MySQL/SQL Server) • Python (Pandas, NumPy, Matplotlib, Seaborn) • Azure Cloud • Azure Data Factory • Power BI • Excel • Git (optional, for portfolio)